

FINAL TEST (Advanced Machine Learning) group B

2025-06-06

- Upload files with solutions and plots to Assignment on MS TEAMS: `test.ipynb` (or `.py` / `.r` with additional `PLOT[plot_number].jpg`).
- Deadline is: 9:50
- DO NOT use archive file format such as zip!
- Uploading files after the deadline or using the wrong format will result in reduced points

1 Task 1 (10 points)

- Read the paired data (x_i, y_i) , $i = 1, \dots, n$, from `weights.csv` representing weights of some person (some dates are missing).
- Randomly split the n observations into four near equal-sized folds $\mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3, \mathcal{F}_4$.
- Perform 4-fold CV for Nadaraya-Watson kernel regression:
 - Choose a set of bandwidths $\{h_1, \dots, h_m\}$.
 - For each h_j , perform a 4-fold CV.
 - Plot MSE vs h .
 - Select the bandwidth h^* that minimizes CV-MSE.
- Perform 4-fold CV for smoothing-splines regression:
 - Choose a set of smoothing parameters $\{\lambda_1, \dots, \lambda_p\}$.
 - For each λ_ℓ , fit the smoothing spline on three folds, compute MSE on the held-out fold, and average over all four folds.
 - Plot MSE vs λ .
 - Select the smoothing parameter λ^* that minimizes CV-MSE.
- Plot the scatter of (x_i, y_i) and overlay the fitted curves $\hat{f}_{h^*}(x)$ and $\tilde{f}_{\lambda^*}(x)$ on a fine grid over the range of x .

2 Task 2 (4 points)

Generate artificial data with as follows:

- Generate feature X_1 from the mixture of 3 Gaussian distributions:

$$0.25 \cdot N(-1, sd = 0.2) + 0.25 \cdot N(1, sd = 0.2) + 0.5 \cdot N(5, sd = 0.2)$$

- Generate features $X_2, \dots, X_{10}$ from $N(0, sd = 1)$.
- Generate binary class variable such that $P(Y = 1|X) = 0.1$ when $X_1 < 3$ and $P(Y = 1|X) = 0.9$ when $X_1 \geq 3$.
- Generate training data $D_{train} = \{(x_i, y_i) : i = 1, \dots, n\}$ of size $n = 1000$, where x_i is 10-dimensional feature vector for i -th observation and y_i is the value of class variable for i -th observation.
- In addition generate testing data D_{test} of size $n = 1000$ using the above scheme.

3 Task 3 (16 points)

Assume that our model predicts $P(Y = 1|X = x)$ using logistic sigmoid function $\sigma(x^T \theta)$ and the optimal values of parameters θ are found by minimizing the following weighted risk function on training data:

$$R(\theta) = -\frac{1}{n} \sum_{i=1}^n \|x_i\|_{\infty} [y_i \cdot x_i^T \theta + \log(1 - \sigma(x_i^T \theta))] + 0.01 \|\theta\|_2^2,$$

To make a prediction for new observation we apply the rule $\hat{y} = 1$ when $\sigma(x^T \hat{\theta}) > 0.5$.

- Calculate the gradient $\nabla R(\theta)$ of the risk function. Write the solution in the comment.
- Implement from scratch ¹ and run gradient descent (GD) algorithm with learning rate $lr = 0.01$ and number of epochs = 100. Take as the initialization point $\theta = (0, \dots, 0)$.
- Generate plot showing how the value of the risk changes for subsequent iterations. Save the results in file [RISK.pdf](#).
- Generate a plot showing how the accuracy of the model calculated on training data changes for subsequent iterations. Save the results in file [TRAIN.pdf](#). Consider also the prediction for the initialization point.
- Generate a plot showing how the accuracy of the model calculated on testing data changes for subsequent iterations. Save the results in file [TEST.pdf](#). Consider also the prediction for the initialization point.

¹It is not allowed to use ready-made implementations of the GD algorithm, e.g. the one from the pytorch library.